

Age Differences in the Online Integration of Sticker-Based Emotional Context in Digital Communication

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1. Introduction

Q1. Emoticon vs Emoji vs K-Sticker

Emoticon	Emoji	K-Sticker
- Typographic - Inline with text - Free	- Unicode-based pictogram - Inline with text - Free	- Large image size & animation - Standalone message unit - Free or paid

Q2. Same Sticker, Same Feeling?

20s	50+
- Amusement - Joy - Enjoyment - Laughter	- Sadness - Embarrassment - Excitement - Disappointment

Background

- Digital communication requires the integration of textual messages with nonverbal visual cues. [1]
- The interpretation of visual cues such as emojis can vary across age groups. [2]
- Socioemotional Selectivity Theory (SST) suggests that older adults prioritize emotionally meaningful information, particularly positive information. [3,4]
- We examined whether age differences in sticker evaluation are linked to real-time processing during conversation.

Research Questions

- RQ1** Do older adults evaluate stickers more positively than younger adults?
- RQ2** Do these age differences vary by sticker category and polarity?
- RQ3** Do affective evaluations predict online processing differently by age?

2. Experiment 1 – Sticker Ratings

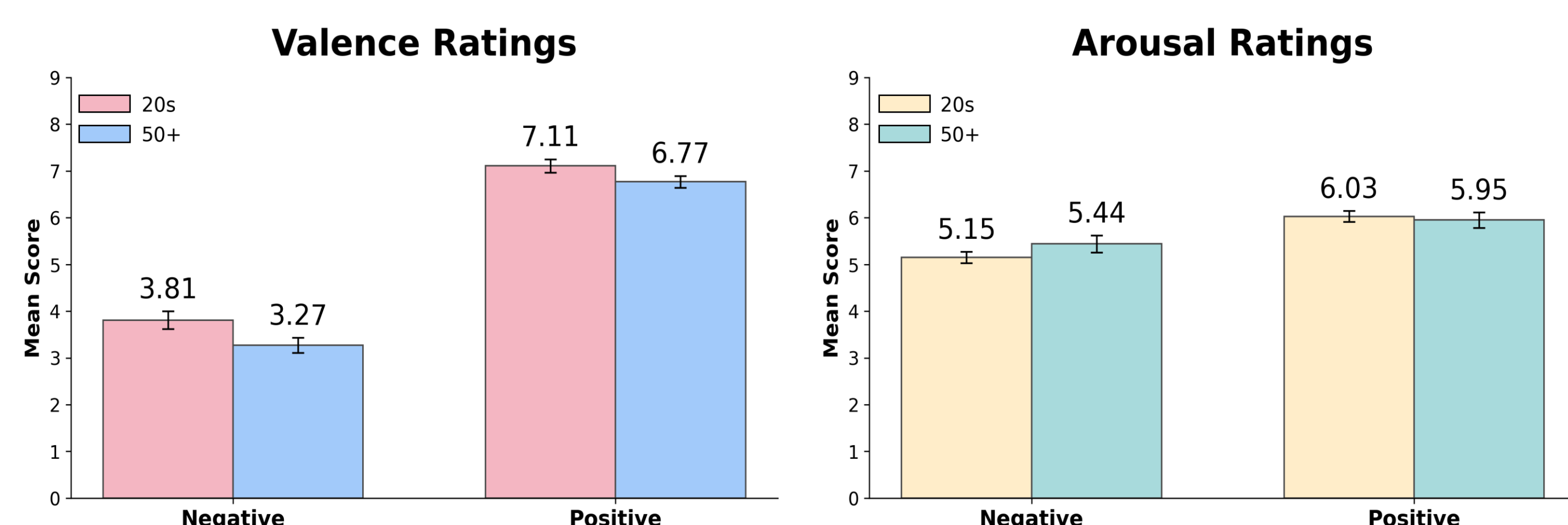
Method

- Participants:** $N=88$ (20s: $n=45$; 50+: $n=43$)
- Task:** Valence and Arousal Questionnaire (9-point scales)
- Examples of Sticker Stimuli**

Session	A		B			
Sticker category	Default		20s-favored		50+-favored	
Polarity	Positive	Negative	Positive	Negative	Positive	Negative
Example						

Session A

- 30 default (free) stickers** — 15 positive, 15 negative
- Analysis:** LMMs — Age $\times$ Polarity (random intercepts for Participant and Item)



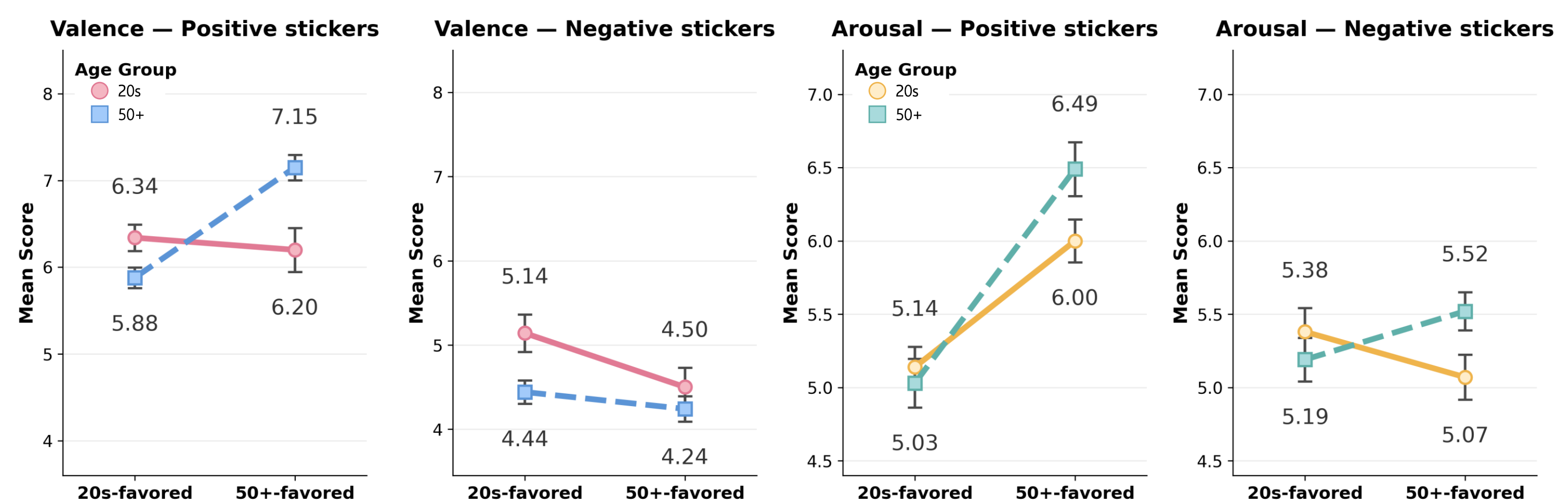
- Valence:** The 50+ group rated default stickers less positively overall (Age: $F(1, 86) = 7.18, p = .009$), and this age difference did not vary by polarity (Age $\times$ Polarity: $F(1, 2522) = 2.97, p = .085$).
- Arousal:** The effect of polarity on arousal differed by age (Age $\times$ Polarity: $F(1, 2522) = 7.68, p = .006$).

⇒ Is there no positivity effect in older adults' sticker evaluations?

2. Experiment 1 – Sticker Ratings

Session B

- 24 stickers most favored by each age group** — 12 favored by the 20s group and 12 by the 50+ group (6 positive, 6 negative each)
- Analysis:** LMMs — Age $\times$ Polarity $\times$ Sticker Category (random intercepts for Participant and Item)



- Valence & Arousal:** Age differences varied by sticker category; for valence, this pattern also varied by polarity (Valence: $F(1, 1998) = 11.20, p < .001$, Arousal: $F(1, 1998) = 19.30, p < .001$).
- For stickers most favored by the 50+ group, 50+ participants gave higher valence ratings for positive stickers and higher arousal ratings overall.

⇒ Are individual differences in affective sticker evaluations reflected in real-time conversational processing?

3. Experiment 2 – Eye Tracking

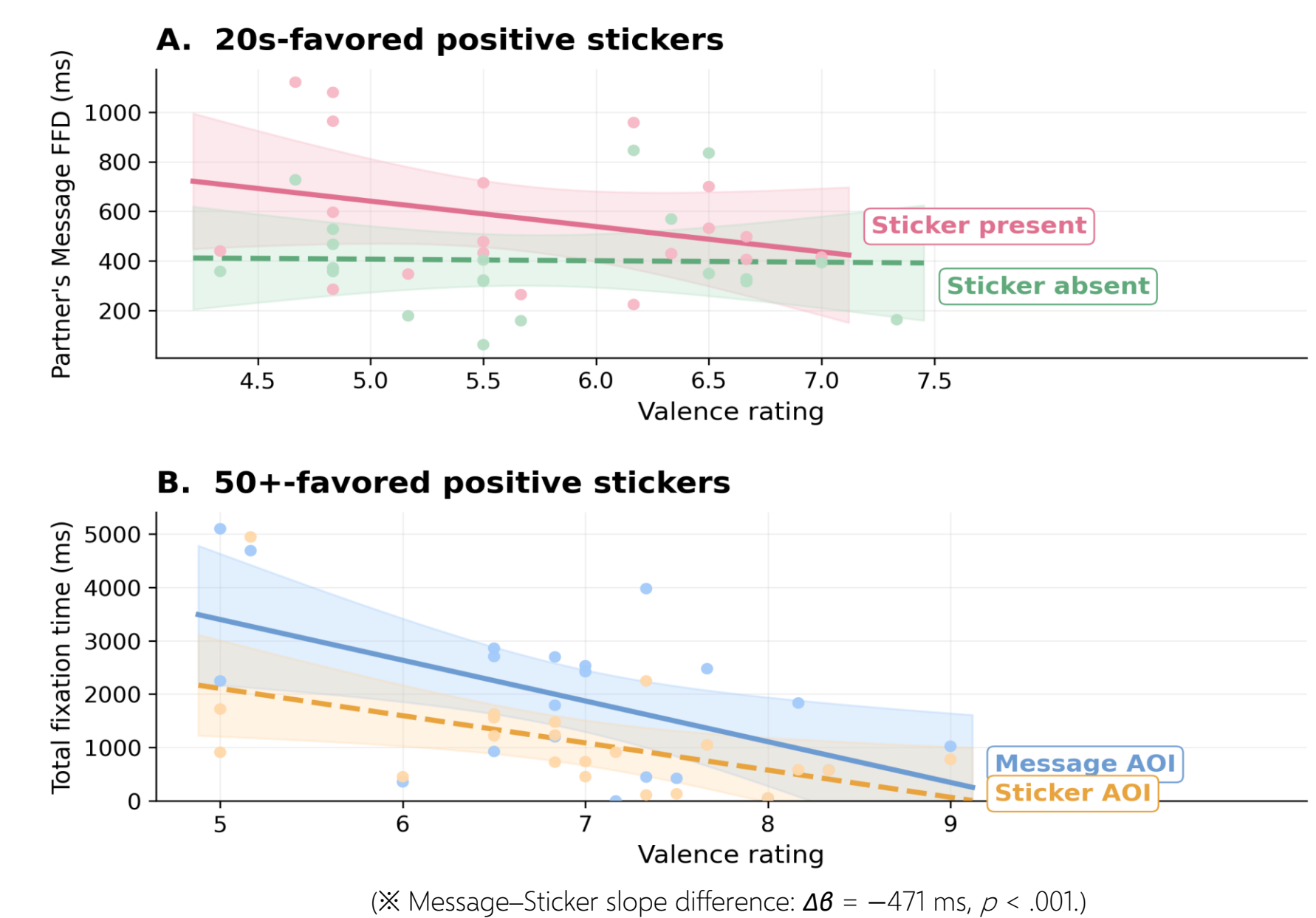
Method

- Participants:** $N=40$ (20 matched pairs; 20s: $n=18$; 50+: $n=22$)
- Task:** Scenario-based dyadic conversation on KakaoTalk for PC.
- Analysis:** LMMs predicting turn-level eye-tracking measures from participant-level Session B valence ratings (Polarity $\times$ Sticker Category), controlling for message length, with random intercepts for Participant and Item.

Chat Interface AOI



FFD: Duration of the first fixation within the Partner's Message AOI.
FT: Sum of all fixation durations within the Message, Sticker, or Combined AOI.



Results

- Follow-up analyses within the 50+ group showed significant associations between positive-sticker ratings and fixation measures.

A. 20s-favored positive stickers — Message FFD

- Higher valence ratings were associated with shorter Partner's Message FFD only when a sticker was present (Age $\times$ Rating $\times$ Sticker Presence: $F(1, 746) = 7.97, p = .005$).

B. 50+-favored positive stickers — Total Fixation Time

- Higher valence ratings were associated with shorter fixation times, with a stronger association for the Message AOI than for the Sticker AOI (Age $\times$ Rating $\times$ AOI: $F(1, 905.59) = 9.08, p = .003$).

Conclusion

- The 50+ group evaluated positive stickers favored by their own age group more positively, consistent with SST's positivity effect.
- In the 50+ group, higher positive-sticker ratings were associated with shorter fixation times during real-time conversation.
- Overall, in the 50+ group, affective ratings varied across age-favored sticker categories, while individual valence ratings for positive stickers were also associated with real-time message processing.

References

- [1] Barach, E., Feldman, L. B., & Sheridan, H. (2021). Are emojis processed like words? Eye movements reveal the time course of semantic processing for emoji-fied text. *Psychonomic Bulletin & Review*, 28(3), 978–991.
- [2] Chen, Y., Yang, X., Howman, H., & Filik, R. (2024). Individual differences in emoji comprehension: Gender, age, and culture. *PLOS ONE*, 19(2), e0297379.
- [3] Carstensen, L. L., Isaacowitz, D. M., & Charles, S. T. (1999). Taking time seriously: A theory of socioemotional selectivity. *American Psychologist*, 54(3), 165–181.
- [4] Mather, M., & Carstensen, L. L. (2005). Aging and motivated cognition: The positivity effect in attention and memory. *Trends in Cognitive Sciences*, 9(10), 496–502.